Unit 1 :: Vector Calculus

PHY-02-MJ-MN-01-R-04: Mathematical Physics I

Dr. Parag Bhattacharya
Department of Physics
Rangapara College (Autonomous)

Syllabus

Unit I: Vector calculus

A. Review of vector algebra: addition and resolution of vectors; dot and cross products; scalar triple product; Cartesian components of a vector; conditions for perpendicularity and collinearity; scalar and vector fields.

B. The del ($\boldsymbol{\nabla}$) operator, gradient of scalar fields; normal and directional derivatives; divergence and curl of vector fields; geometrical interpretation of gradient, divergence and curl operations; the Laplacian (∇^2) operator; vector differential identities; the Jacobian.

C. Notion of infinitesimal line, surface and volume elements; line, surface and volume integrals; flux of a vector field; Gauss' divergence theorem; Stokes' and Green's theorems (qualitative treatment only); applications of Gauss', Stokes' and Green's theorems.

1 Review of vector algebra

1.1 Addition of vectors

Two or more vectors may be added by geometrical laws, which are discussed below:

1. *Triangle law of vector addition*

   Statement: If two vectors are represented both in magnitude and direction by two sides of a triangle taken in the *same order*, then their resultant vector is represented both in magnitude and direction by the third side of the triangle taken in the *opposite order*.

   Note: Vectors are in the *same order* when one's origin coincides with the other's terminus; otherwise, they are in the *opposite order*.

2. *Parallelogram law of vector addition*

   Statement: If two vectors acting at the same point are represented both in magnitude and direction by the adjacent sides of a parallelogram, then their resultant is represented both in magnitude and direction by the diagonal of that parallelogram passing through the same point.

3. *Polygon law of vector addition*

   Statement: If several vectors are represented both in magnitude and direction by the sides of a polygon taken in order, then their resultant is represented both in magnitude and direction by the closing side taken in reverse order.

1.1.1 Properties of vector addition

i. Vector addition is *commutative*, i.e.

$$\vec{\boldsymbol{A}}+\vec{\boldsymbol{B}}=\vec{\boldsymbol{B}}+\vec{\boldsymbol{A}}$$

ii. Vector addition is *associative*, i.e.

$$\vec{\boldsymbol{A}}+\left(\vec{\boldsymbol{B}}+\vec{\boldsymbol{C}}\right)=\left(\vec{\boldsymbol{A}}+\vec{\boldsymbol{B}}\right)+\vec{\boldsymbol{C}}$$

iii. Vector addition is *distributive*, i.e.

$$\lambda\left(\vec{\boldsymbol{A}}+\vec{\boldsymbol{B}}\right)=\lambda\vec{\boldsymbol{A}}+\lambda\vec{\boldsymbol{B}}$$

where λ is a real number.

iv. The sum of two negative vectors is a null vector. So if vectors $\vec{\boldsymbol{A}}$ and $\vec{\boldsymbol{B}}$ are negatives of each other, i.e.

$$\vec{\boldsymbol{A}}=-\vec{\boldsymbol{B}}$$

then we have

$$\vec{\boldsymbol{A}}+\vec{\boldsymbol{B}}=\vec{\boldsymbol{0}}$$

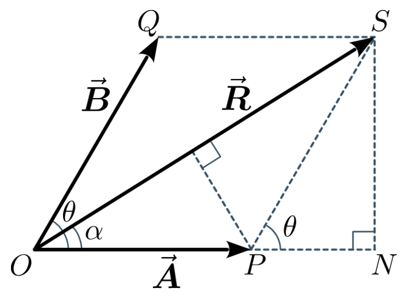


Figure 1: Magnitude of a vector sum

1.1.2 Magnitude of the vector sum

Given two vectors $\vec{A}$ and $\vec{B}$ and their resultant $\vec{R} = \vec{A} + \vec{B}$ as per the parallelogram law. Consider the construction as given in figure 1. Because now that $OPSQ$ is a parallelogram, we have

$$\begin{aligned} OP &= QS = |\vec{A}| = A \\ OQ &= SP = |\vec{B}| = B \\ \text{and} \quad OS &= |\vec{R}| = R \end{aligned}$$

Let θ be the angle between vectors $\vec{A}$ and $\vec{B}$. Then we have $\angle SPN = \theta$. Hence, from $\triangle PSN$, we obtain

$$\begin{aligned} PN &= SP\cos\theta = B\cos\theta \\ \text{and} \quad SN &= SP\sin\theta = B\sin\theta \end{aligned}$$

Then in $\triangle OSN$, using Pythagoras' theorem, we get

$$\begin{aligned} OS^2 &= ON^2 + SN^2 \\ \implies OS^2 &= (OP + PN)^2 + SN^2 \\ \implies R^2 &= (A + B\cos\theta)^2 + (B\sin\theta)^2 \\ &= A^2 + B^2\cos^2\theta + 2AB\cos\theta + B^2\sin^2\theta \\ &= A^2 + B^2\left(\cos^2\theta + \sin^2\theta\right) + 2AB\cos\theta \\ \implies R^2 &= A^2 + B^2 + 2AB\cos\theta \end{aligned}$$

Therefore,

$$\boxed{|\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}}$$

1.1.3 Direction of the vector sum

Let α be the angle between the resultant vector $\vec{R} = \vec{A} + \vec{B}$. Because direction of the vector $\vec{A}$ is already known, knowing the angle α would enable us to know the direction of $\vec{R}$ with respect to $\vec{A}$.

From $\triangle OSN$, we obtain,

$$\begin{aligned} \tan\alpha &= \frac{SN}{ON} \\ &= \frac{SN}{OP+PN} \\ \implies \tan\alpha &= \frac{B\sin\theta}{A+B\cos\theta} \end{aligned}$$

Therefore,

$$\boxed{\alpha = \tan^{-1}\left(\frac{B\sin\theta}{A+B\cos\theta}\right)}$$

1.1.4 Subtraction on vectors

The subtraction of a vector from another may be defined in terms of the addition of one vector with the negative of the other vector, i.e.

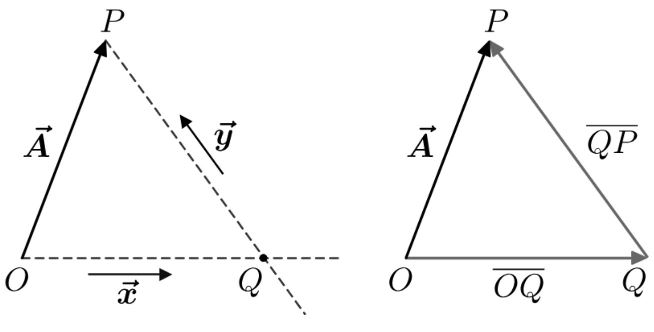
$$\vec{\boldsymbol{A}} - \vec{\boldsymbol{B}} = \vec{\boldsymbol{A}} + \left(-\vec{\boldsymbol{B}}\right)$$

The magnitude of the vector difference is:

$$\boxed{|\vec{\boldsymbol{A}} - \vec{\boldsymbol{B}}| = \sqrt{A^2 + B^2 - 2AB\cos\theta}}$$

1.1.5 Resolution of vectors

Suppose we have a certain vector $\vec{\boldsymbol{A}}$ and two arbitrary vectors $\vec{\boldsymbol{x}}$ and $\vec{\boldsymbol{y}}$, all of them being coplanar. Let us now perform the following geometrical constructions.



I. Through the origin O of vector $\vec{\boldsymbol{A}}$, we draw a straight line parallel to $\vec{\boldsymbol{x}}$.

II. Through the terminus P of vector $\vec{\boldsymbol{A}}$, we draw another straight line parallel to $\vec{\boldsymbol{y}}$.

III. Let the point of intersection of these two straight lines be Q.

IV. We define the following two vectors:

$$\begin{aligned} \overline{OQ}: &\quad \text{origin at } O\text{, terminus at } Q \\ \overline{QP}: &\quad \text{origin at } Q\text{, terminus at } P \end{aligned}$$

Then, by triangle law of vector addition, we have

$$\vec{\boldsymbol{A}} = \overline{OQ} + \overline{QP} \tag{1}$$

We know that when a vector is multiplied by some scalar, we obtain another parallel vector with different magnitude. Conversely, for two parallel vectors, one of them can be expressed as a scalar multiple of the other.

Consequently, because $\overline{OQ}$ is parallel to $\vec{\boldsymbol{x}}$, we have

$$\overline{OQ} = \alpha\vec{\boldsymbol{x}} \tag{2}$$

Also, because $\overline{QP}$ is parallel to $\vec{\boldsymbol{y}}$, we have

$$\overline{QP} = \beta\vec{\boldsymbol{y}} \tag{3}$$

where α, β are some arbitrary constant scalars.

Using equations (2) and (3) in equation (1) gives

$$\vec{\boldsymbol{A}} = \alpha\vec{\boldsymbol{x}} + \beta\vec{\boldsymbol{y}} \tag{4}$$

Therefore, equation (4) shows that given some specific vector, it can be expressed as a vector sum of two arbitrary vectors, with each being multiplied by some constant scalars, and all of them lying on a plane.

Then we may say that the given vector has been resolved along two arbitrary vectors on the same plane, and the procedure followed to do so is known as the *resolution of a vector*.

1.2 Multiplication of vectors

1.2.1 Scalar product (or dot product)

Given two vectors $\vec{\boldsymbol{A}}$ and $\vec{\boldsymbol{B}}$ which have an angle θ between them, their scalar product is a scalar quantity which equals the product of their magnitudes and the cosine of the angle between them.

It is expressed mathematically as

$$\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}} = |\vec{\boldsymbol{A}}||\vec{\boldsymbol{B}}|\cos\theta$$

$$\text{or,}\quad \boxed{\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}} = AB\cos\theta}$$

where $A = |\vec{\boldsymbol{A}}|$ and $B = |\vec{\boldsymbol{B}}|$.

Some of the properties of the dot product are:

1. *Commutativity*: The order of vectors does not matter, i.e.

$$\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}} = \vec{\boldsymbol{B}} \cdot \vec{\boldsymbol{A}}$$

2. *Distributivity*: The scalar product distributes over vector addition, i.e.

$$\vec{\boldsymbol{A}} \cdot \left(\vec{\boldsymbol{B}} + \vec{\boldsymbol{C}}\right) = \vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}} + \vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{C}}$$

3. The scalar product of a vector with itself equals the square of its magnitude, i.e.

$$\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{A}} = A^2$$

4. Multiplying either of the vectors by a scalar affects the scalar product by the same factor, i.e.

$$\left(\lambda\vec{\boldsymbol{A}}\right) \cdot \vec{\boldsymbol{B}} = \vec{\boldsymbol{A}} \cdot \left(\lambda\vec{\boldsymbol{B}}\right) = \lambda\left(\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}}\right)$$

1.2.2 Vector product (or cross product)

Given two vectors $\vec{\boldsymbol{A}}$ and $\vec{\boldsymbol{B}}$ which have an angle θ between them, their vector product is another vector quantity which is perpendicular to the plane that contains both $\vec{\boldsymbol{A}}$ and $\vec{\boldsymbol{B}}$, and is denoted by $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}$.

– The magnitude of $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}$ is obtained as

$$|\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}| = AB\sin\theta$$

– The direction of $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}$ is obtained using the right-hand thumb rule, which is applied as follows:

> If the fingers of one's right hand are curled from the first vector to the second, then the stretched thumb of the right hand points in the direction of their cross product.

– The vector product is anti-commutative, i.e.

$$\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = -\vec{\boldsymbol{B}} \times \vec{\boldsymbol{A}}$$

Because $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}$ and $\vec{\boldsymbol{B}} \times \vec{\boldsymbol{A}}$ are negatives of each other, the order of the cross product is important.

Some of the properties of the cross product are:

1. *Anti-commutativity*: The order of vectors is important, and reversing it changes the direction of the vector product, i.e.

$$\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = -\vec{\boldsymbol{B}} \times \vec{\boldsymbol{A}}$$

2. *Perpendicularity*: The resultant vector product is perpendicular to the plane containing the original vectors.

3. *Magnitude*: The magnitude of the vector product is the area of the parallelogram formed by the two vectors.

4. *Distributivity*: The vector product distributes over vector addition, i.e.

$$\vec{\boldsymbol{A}} \times \left(\vec{\boldsymbol{B}} + \vec{\boldsymbol{C}}\right) = \vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} + \vec{\boldsymbol{A}} \times \vec{\boldsymbol{C}}$$

1.2.3 Scalar triple product

Given three vectors $\vec{\boldsymbol{A}}$, $\vec{\boldsymbol{B}}$ and $\vec{\boldsymbol{C}}$, their scalar triple product is given as

$$[\vec{\boldsymbol{A}}\vec{\boldsymbol{B}}\vec{\boldsymbol{C}}] = \vec{\boldsymbol{A}} \cdot \left(\vec{\boldsymbol{B}} \times \vec{\boldsymbol{C}}\right)$$

It is calculated as the determinant of a 3×3 matrix formed by the component vectors, with a result of zero indicating the vectors are coplanar.

- The operation combines a dot product and a cross product acting on three vectors to produce a scalar quantity.